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OCTOBER 8, 1948
SUPERSEDING
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June 8, 1938

FEDERAL SPECIFICATION

BRUSHES, CLEANING; LABORATORY GLASSWARE

This specification was approved by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government.

1. CLASSIFICATION

1.1 Types, styles, and sizes.—Laboratory glassware cleaning brushes covered by this specification shall be of the following types, styles, and sizes, as specified: (See figs. 1 to 6, inclusive.)

Type I.—Beaker.
Type II.—Bottle and burette.
Sizes.—½, ¾, 1, 1¼, 1½, 2, 2½.
Type III.—Cylinder.
Type IV.—Pipette.
Type V.—Test-tube.
Style A.—Dome end.
Sizes.—½, ½XL, ¾, ¾XL, 1, 1¼, 1½, 1¾, 2, 2¼.
Style B.—Tuft end.
Sizes.—½, ¾, 1, 1¼, 1½, 1¾, 2, 2¼.
Type VI.—Open-end tube.

2. APPLICABLE SPECIFICATIONS

2.1 Specifications.—There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Standard sample.—The brushes shall be equal to the standard sample in workmanship and balance; in stiffness, density and kind of bristle, and in any other respects not specifically covered in this specification. (See 6.1 and 6.2.)

3.2 Designs.—The brushes shall be of the general designs illustrated in figures 1 to 6, inclusive.

3.3 Material.

3.3.1 *Bristle.*—The bristle shall be Chinese Chungking of uniform length, trimmed so as to show no tendency to separate at the ends, and shall have a good spring.

3.3.2 *Wire.*—The wire shall be brass or shall be steel made corrosion-resistant by zinc or other suitable coating.

3.3.3 *Wooden handles.*—The wooden handles shall be of close-grained hardwood, and shall have a smooth, natural, varnished, lacquered, or enameled finish.

3.4 Type I, beaker brushes.

3.4.1 *Construction.*—The bristles shall be secured within two loops of twisted wires having four ends firmly attached to a wooden handle. The bristles shall be approximately parallel to each other and at an angle of approximately 90° to the wire. The bristles shall be evenly distributed in the brush head. (See fig. 1.)

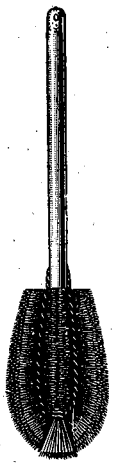


FIGURE 1.—Type I, beaker brush.

3.4.2 *Tufts.*—Two stiff but resilient Chungking bristle tufts shall be attached at the end of the brush head. The tufts shall be neatly and

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securely bound near their bases with wire and shall extend at least ⅜ inch beyond the end of the brush bristle.

3.4.3 *Handle.*—The handle shall be round. There shall be a hole approximately 5/16 inch in diameter, centrally located about ¾ inch from the hand end of the brush.

3.4.4 *Physical requirements.*—Type I brushes shall conform to the physical requirements in table I.

TABLE I.—*Physical requirements of type I, beaker brushes*

Length, over-all, inches	16½ ± 1
Brush head:	
Length, including tufts, do	5 ± ¼
Width, do	2¾ ± ¼
Bristle:	
Length, do	1½ ± ⅛
Handle:	
Length, do	16 ± 1
Diameter, inch	½ ± 1/16
Wire (twisted):	
Diameter, single strand, minimum, inch	0. 062

3.5 Type II, bottle and burette brushes.

3.5.1 *Construction.*—The bristles shall be secured within twisted wires which shall also form the handle of the brush. The hand end shall have a loop. The bristles shall be approximately parallel to each other and at an angle of approximately 90° to the wire. The bristles shall be evenly distributed in the brush head. (See fig. 2.)

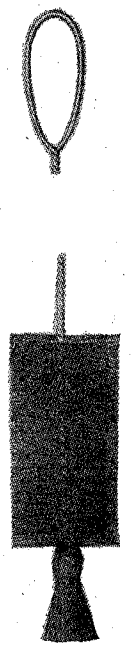


FIGURE 2.—Type II, bottle and burette brush.

3.5.2 *Tuft.*—There shall be a neatly trimmed tuft of bristles extending beyond the brush end, neatly and securely bound with wire.

3.5.3 *Physical requirements.* — Type II brushes shall conform to the physical requirements in table II.

TABLE II.—*Physical requirements of type II, bottle and burette brushes*

	Size						
	½	¾	1	1¼	1½	2	2½
Length, over-all, minimum, inches	36	36	16	20	12	16	22
Brush head:							
Length, minimum (exclusive of tuft), do	5¼	5¼	4½	5	4	4	4
Diameter ± 1/16, do	½	¾	1	1¼	1¼	2	2½
Wire (twisted):							
Diameter, single strand, minimum, inch	0. 062	0. 062	0. 062	0. 080	0. 080	0. 090	0. 090

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3.6 Type III, cylinder brushes.

3.6.1 *Construction.*—The bristles shall be secured within two loops of twisted wires having four ends firmly attached to a wooden handle. The bristles shall be approximately parallel to each other and at an angle of approximately 90° to the wire. The bristles shall be evenly distributed in the brush head. The bristles shall be trimmed to give the brush a conical shape. (See fig. 3.)

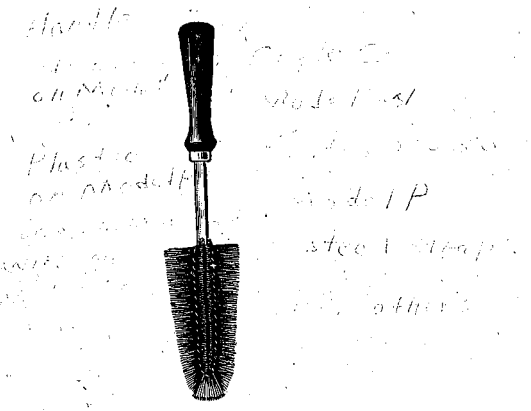


FIGURE 3.—Type III, cylinder brush.

3.6.2 *Tufts.*—Two stiff but resilient bristle tufts shall be attached to the end of the brush head. Tufts shall be neatly and securely bound near their bases with wire and shall extend at least ⅜ inch beyond the end of bristles of the brush head.

3.6.3 *Metal bar.*—The metal bar shall be rounded at the edges. The bar shall be not less than 0.31 inch in width and not less than 0.09 inch in thickness. The bar shall be glue-set into the handle.

3.6.4 *Handle.*—The handle shall be rounded and shaped to fit the hand. There shall be a hole approximately 3/16 inch in diameter, centrally located about ¾ inch from the hand end of the brush.

3.6.5 *Ferrule.*—There shall be a corrosion-resistant metal ferrule approximately ⅝ inch wide clamped around the handle where the bar is inserted into the wooden handle.

3.6.6 *Physical requirements.* — Type III brushes shall conform to the physical requirements in table III.

TABLE III.—*Physical requirements of type III, cylinder brushes*

Length, over-all, inches	12 ± ½
Brush head:	
Length, including tufts, do	5 ± ⅛
Diameter at top, do	2¾ ± ¼
Bristle:	
Length, do	1½ ± ⅛
Handle:	
Length, do	4½ ± ½
Diameter, inch	¾ ± ¼
Wire (twisted):	
Diameter, single strand, minimum inch	0.062

3.7 Type IV, pipette brushes.

3.7.1 *Construction.*—The bristles shall be secured within twisted wires which shall also form the handle of the brush. The hand end shall have a loop. The bristles shall be approximately parallel to to each other and at an angle of approximately 90° to the wire. The bristles shall be evenly distributed in the brush head and shall be shaped as shown in figure 4.



FIGURE 4.—Type IV, pipette brush.

3.7.2 *Physical requirements.* — Type IV brushes shall conform to the physical requirements in table IV.

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TABLE IV.—*Physical requirements of type IV, pipette brushes*

Length, over-all, inches	17 ± ½
Brush head:	
Length, do	7 ± ⅛
Diameter, top 3 inches, inch	⅝ ± 1/16
Diameter at bottom, do	¼ ± 1/16
Wire (twisted):	
Diameter, single strand, minimum, inch	0.031

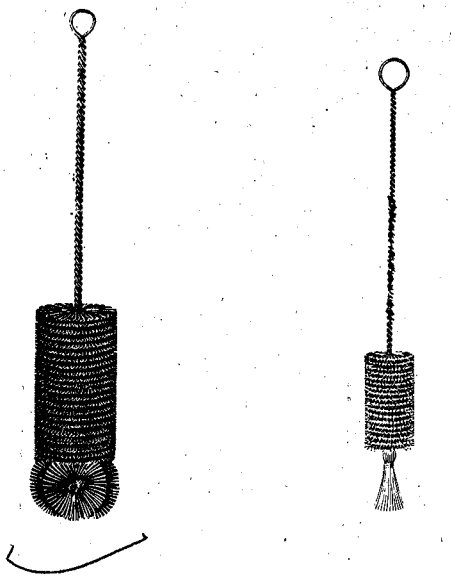
3.8 Type V, test-tube brushes.

3.8.1 *Style A, dome end.*

3.8.1.1 *Construction.*—The bristles of style A brushes shall be secured within twisted wires which shall also form the handle of the brush. The hand end shall have a loop. The bristles shall be approximately parallel to each other and at an angle of approximately 90° to the wire. The bristles shall be evenly distributed in the brush head. The lower end of the brush part shall have a dome composed of several fan-shaped wings, neatly and securely bound into the twisted wire. The dome shall extend at least ½ inch beyond the ends of the bristles of the brush head. (See fig. 5.)

3.8.2 *Style B, tuft end.*

3.8.2.1 *Construction.*—The bristles of style B brushes shall be secured within twisted wires which also form the handle of the brush. The hand end shall have a loop. The bristles shall



STYLE A STYLE B

FIGURE 5.—Type V, test-tube brushes.

TABLE V.—*Physical requirements of type V, test-tube brushes*

	Size									
	½	½XL	¾	¾XL	1	1¼	1½	1¾	2	2¼
STYLE A										
Length, over-all, minimum, inches	10	30	10	36	10	10	10	12	15	20
Brush head:										
Length exclusive of dome, minimum, inches	2½	3¼	3	3¼	3¼	3½	3½	4½	4½	4½
Diameter ± 1/16, do	½	½	¾	¾	1	1¼	1½	1¾	2	2¼
Wire (twisted):										
Diameter, single strand, minimum, inch	0.062	0.062	0.062	0.062	0.062	0.080	0.080	0.080	0.090	0.090
STYLE B										
Length, over-all, minimum, inches	9	------	9	------	9	9	9	9	9	9
Brush head:										
Length including tuft, minimum, inches	3½	------	3½	------	3½	3½	3½	3½	3½	3½
Diameter ± 1/16, do	½	------	¾	------	1	1¼	1½	1¾	2	2¼
Wire (twisted):										
Diameter, single strand, minimum, inch	0.062	------	0.062	------	0.062	0.080	0.080	0.080	0.090	0.090

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be approximately parallel to each other and at an angle of approximately 90° to the wire. The bristles shall be evenly distributed in the brush head. (See fig. 5.)

• **3.8.2.2** *Tuft.*—There shall be a neatly trimmed tuft of bristles extending beyond the brush end, neatly and securely bound with wire.

3.8.3 *Physical requirements.*—Type V brushes shall conform to the physical requirements in table V.

3.9 Type VI, open-end tube brushes.

3.9.1 *Construction.*—The bristles shall be secured within twisted wires which shall also form the handle of the brush. The handle end of the wires shall have a loop; the other end of the wires shall be tightly twisted to a rounded, smooth end. The bristles shall be centrally located in the middle of the length of the twisted wire. (See fig. 6.)

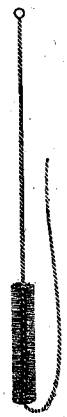


FIGURE 6.—Type VI, open-end tube brush.

3.9.2 *Physical requirements.*—Type VI brushes shall conform to the physical requirements in table VI.

TABLE VI.—*Physical requirements of type VI, open-end tube brushes*

Length, over-all—inches	24±½
Brush part:	
Length—do	3±⅛
Diameter—inch	½±1/16
Wire (twisted):	
Diameter, single strand, minimum—inch	0.031

3.10 Workmanship.—The brushes shall be free from defects which affect their appearance or which may affect their serviceability. Brushes and their component parts shall be neatly finished, fitted, and assembled, and shall have no sharp edges.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Two brushes shall be taken from each lot of 100 brushes or fraction thereof delivered.

4.2 Inspection.

4.2.1 The brushes shall be inspected to determine compliance with this specification with respect to material, construction, and workmanship; size and finish of handles; diameter and finish of wire; length of bristle; overall length of brush; and dimensions of brush head.

4.2.2 The brushes shall be compared with the standard sample with respect to the requirements of 3.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, commercial packages are acceptable.

5.2 Packing.—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Issue packages.*—Unless otherwise specified, issue packages shall be marked with the name of the material, the type, and if applicable, the style and size, the quantity contained therein, and the names of the contractor and manufacturer.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the type, and if applicable, the style and size, as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Ordering information. — Purchasers should specify the type, and in addition, for type II, the size, and for type V, the style and

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size, and the location at which standard samples may be seen, and should exercise any desired options offered herein. (See 1.1, 3.1, 5.1, 5.3, and 7.4.5.)

6.2 Standard sample.—The standard sample is a reasonable facsimile of the master standard sample on file at the National Bureau of Standards, Washington 25, D. C. It is not intended that the master standard sample shall be used for the purpose of purchase or inspection of brushes, but that the master standard sample will be used by Government agencies for comparison purposes to obtain a standard sample for their own use in purchase and inspection.

6.2.1 In order that uniform standards for brushes may be maintained among the various Government agencies, purchasers should forward brushes to the National Bureau of Standards for inspection and its opinion in order to obtain a standard sample equal to the master standard sample; the standard sample should then be preserved by the purchaser and used in the purchase and inspection of brushes obtained under this specification.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army.

7.2.1 *Applicable specifications.*

Federal Specifications:

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

Joint Army-Navy Specifications:

JAN–P–103—Packaging and Packing for Overseas Shipment—Boxes; Wood, Cleated; Solid Fiberboard.

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN–P–133—Packaging and Packing for Overseas Shipment—Boxes, Set-up, Paperboard.

JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

U. S. Army Specification:

100–2—Standard Specification for Marking Shipments by Contractors.

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.)

7.2.2 *Packaging.*

7.2.2.1 Each brush shall be individually wrapped in two plys of 30-pound basis weight kraft paper based on ream of 24 x 36—500. The ends of the kraft paper shall be folded back over each end of the brush and secured with strips of gummed paper tape.

7.2.2.2 Twelve brushes shall be packed in full or partial telescope boxes which conform to the requirements of Joint Army-Navy Specification JAN–P–133, type I or II, class A. Brushes shall be arranged in a reversing end-to-end order. Each box shall contain at least three balls of naphthalene or equivalent quantity of flake naphthalene or other suitable moth repellent to protect the contents from moths.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment.*—Unless otherwise specified, brushes packaged as described in 7.2.2, shall be packed 12 units in corrugated or solid fiberboard boxes which shall be fabricated

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in accordance with Federal Specification LLL–B–631 or LLL–B–636, except that the boxes shall conform to the following:

Gross weight, maximum	Average bursting strength, minimum	Sum of inside dimensions, maximum
Pounds	*P. s. i.*	*Inches*
40	200	75
65	275	90
90	350	100

All flaps of the fiberboard box shall be securely sealed with a good quality adhesive applied throughout the entire area of contact between the flaps.

7.2.3.2 *For overseas shipment.*—Unless otherwise specified, brushes packed as described in 7.2.3.1, shall be overpacked in shipping containers constructed and strapped to conform to the requirements of Joint Army-Navy Specification JAN–P–103, JAN–P–105, JAN–P–106, style 2 or 4, or JAN–B–107. Each shipping container shall be lined with a sealed case liner or prefabricated bag made from waterproof barrier material conforming to the requirements for type L–2 or M of Joint Army-Navy Specification JAN–P–125 and sealed with adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140. A minimum 2-inch overlap shall be provided at all overlapping edges, seams, joints, or closures.

7.2.4 *Marking.*

7.2.4.1 *Packages.*—Each intermediate set-up box shall have a white paper label approximately 2 inches by 4 inches securely glued to one end of box top. The label shall be legibly printed or stamped with indelible ink in letters not less than ¼ inch in height as follows:

Stock No.
Nomenclature
Quantity
Type
Style and size (where applicable)
Contractor's name

7.2.4.2 *Shipping containers.*—Each shipping container shall be marked to comply with the requirements of U. S. Army Specification 100–2, and in addition, the following information shall be stenciled or printed on the container in the following locations, in the order listed:

One end:
Stock number or other identifying number
Nomenclature in detail
Quantity
Type
Style and size (where applicable)
Weight and cube

One side:
Name of contractor
Contract number
Date packed

This information to appear on a side of the container may be a printed label securely affixed, or may be printed or stenciled directly onto the side of the shipping containers. All letters shall be the same height. Minimum size of lettering shall be ¼ inch in height. Maximum area to be covered by label, or space occupied by printing or stenciling, shall be 42 square inches.

7.3 Air Force.—Shipping containers shall be marked in accordance with U. S. Army Specification 94–40645, Marking; Exterior, Domestic and Export Shipment, by Contractors.

7.3.1 *Air Force purchases.*—Each brush of types I and III shall be permanently marked in a plain and legible manner with the letters "U. S."

(Copies of specifications required for Air Force purchases may be obtained from the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio.)

7.4 Navy.

7.4.1 *Applicable specifications and other publications.*

Federal Specifications:
NN–B–591—Boxes; Wood-Cleated-Fiberboard.
NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.
NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
NN–B–631—Boxes; Wood, Wire-bound, for Domestic Shipment.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

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Joint Army-Navy Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN–P–138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.

JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

Navy Department Specifications:

General Specifications for Inspection of Material.

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.

(Copies of Federal and Joint Army–Navy specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.)

7.4.2 *Packaging.*—Unless otherwise specified, standard packages are acceptable. Each box or package shall contain sufficient ball or flake naphthalene to protect the contents from moths.

7.4.3 *Packing.*

7.4.3.1 *For domestic shipment.*—Cleaning brushes, packaged as described in 7.4.2, shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to the point of delivery. When wood-cleated fiberboard boxes, wood-cleated plywood boxes, nailed wood boxes, wire-bound wood boxes, corrugated fiber boxes, or solid fiber boxes are used, they shall conform to the requirements of Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, and LLL–B–636, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

7.4.3.2 *For overseas shipment.* — Cleaning brushes, packaged as described in 7.4.2, shall be packed in cleated plywood boxes, nailed wood boxes, wire-bound wood boxes, fiberboard boxes, or fiberboard-lined wood boxes conforming to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, JAN–B–107, JAN–P–108, and JAN–P–138, respectively. Shipping containers shall be lined with a sealed waterproof bag, or its equivalent, made from barrier material conforming to the requirements of Joint Army-Navy Specification JAN–P–125. The seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

7.4.4 *Marking.*—In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with the requirements of the Navy Shipment Marking Handbook.

7.4.5 *Ordering information.*—Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment. (See 7.4.3 and 7.4.4.)

NAVY: MCMOShS

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog

For sale by the Superintendent of Documents, Washington 25, D. C. Price cents

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